IEEE Access
Multidisciplinary ⋮ Rapid Review ⋮ Open Access Journal

Received 20 February 2025, accepted 7 March 2025, date of publication 13 March 2025, date of current version 21 March 2025.

Digital Object Identifier 10.1109/ACCESS.2025.3550947

RESEARCH ARTICLE

SCL-YOLOv11: A Lightweight Object Detection Network for Low-Illumination Environments

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This work was supported in part by the Research Project of Hainan Vocational University of Science and Technology under Grant HKKY2024-TD-09, in part by the 2025 Hainan Province Higher Education Institutions (Key Scientific Research) Project under Grant Hnky2025ZD-17, and in part by Hainan Provincial Natural Science Foundation of China under Grant 625RC794.

ABSTRACT Object detection in low-light environments has been widely recognized as a critical research direction in the field of computer vision. In response to the challenges of reduced detection accuracy and high edge-deployment costs encountered by mainstream single-stage object detection models under low-light conditions, this paper proposes a lightweight object detection network based on YOLOv11, integrates StarNet, C3k2-Star, and a lightweight detail-enhanced convolution and shared convolutional detection head(LSDECD), so called SCL-YOLOv11 herein. First, the StarNet architecture is introduced into the Backbone to enhance the extraction of shallow image features and significantly reduce computational complexity. Next, Star Blocks from the StarNet framework are employed to optimize the C3k2 module in the Neck stage, thereby improving the localization accuracy of deep features without increasing network complexity. Meanwhile, the Minimum Point Distance based IoU(MPDIoU) loss function is adopted to mitigate gradient explosion risks while enhancing detection precision. Furthermore, a lightweight detail-enhancement convolution layer and a shared-convolution detection head are designed to improve the model's capability in capturing fine-grained details. Finally, a knowledge distillation procedure is conducted using SCL-YOLOv11 as the student model and YOLOv8 as the teacher model, ensuring that the proposed SCL-YOLOv11 model achieves both low complexity and excellent detection performance. Experimental results on the ExDark dataset demonstrate that the proposed algorithm achieves 67.6% mAP@0.5 and 42.4% mAP@0.5:0.95, while reducing parameter count by 38.5% and computational cost by 25.4%. Consequently, the proposed model is highly suitable for edge deployment in industrial applications and embedded development systems.

INDEX TERMS Low-illumination environments, SCL-YOLOv11, lightweight, knowledge distillation.

I. INTRODUCTION

With the rapid development of artificial intelligence, deep learning-based object detection techniques have found widespread applications in various domains, including industry (e.g., intelligent sorting and defect detection), transportation (e.g., autonomous driving), and agriculture (e.g., disease detection) [1]. The primary goal of these techniques is to locate predefined targets in input images and accurately determine their respective classes. Although mainstream single-stage object detection algorithms have achieved remarkable results when dealing with high-quality images captured under sufficient lighting conditions, real-world scenarios often involve poor illumination (e.g., insufficient or partial exposure). Such conditions lead to low contrast between the target and background noise, blurred edges, and loss of detail information, thereby degrading the expressive power of target features [2]. Consequently, the model fails to extract useful information and may generate false positives or miss detections, ultimately resulting in low detection accuracy and imprecise bounding box localization [3].

Existing methods for object detection in low-light conditions can be broadly classified into two categories. The first category adopts a two-stage approach that combines

The associate editor coordinating the review of this manuscript and approving it for publication was Ines Domingues.

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image enhancement and object detection. These methods employ image enhancement algorithms such as RetinexNet [4] or Zero-Reference Deep Curve Estimation(ZeroDCE) [5] to improve image quality and counteract the decline in detection accuracy caused by insufficient illumination. However, because image enhancement and object detection operate independently, computational resources cannot be shared, leading to computational redundancy. Moreover, the two processes are not jointly optimized; thus, the enhanced features may not perfectly align with the requirements of the detection network, adversely affecting the overall system performance. The second category focuses on improving object detection algorithms themselves to enable single-stage detection, wherein the detection model is directly optimized to maintain high accuracy in low-light conditions without relying on additional image enhancement. This category often provides advantages in both efficiency and performance and has become a research hotspot. For example, Mei et al. [6]. addressed the severe performance degradation of YOLO-based models in low-light environments by proposing a novel model—Grouping Offset and Isolated GiraffeDet Target Detection-YOLO (GOI-YOLO)—based on the YOLOv8 architecture. This model primarily employs a repGFPN feature pyramid network to enhance hierarchical feature fusion and integrate low-light information more effectively, thereby ensuring excellent performance on the ExDark dataset. However, this enhancement also increases the parameter count and FLOPs, leading to higher computational costs and memory consumption, which may hinder subsequent deployment and practical applications. In another approach, Han et al. [7]. proposed a YOLOv8n-based 3L-YOLO model to address low detection accuracy under low-light conditions. This model eliminates the need for an image enhancement module, thereby reducing computational costs and achieving satisfactory detection performance on the ExDark dataset. Nonetheless, the reduction in computational cost was insufficient to prevent slow inference speeds. Additionally, Shovo et al. [8]. conducted a comprehensive study using state-of-the-art YOLO models—including YOLOv3, YOLOv5, YOLOv6, and YOLOv8—for low-light object detection, aiming to identify the best-performing YOLO model under such conditions. Their experiments on the ExDark dataset demonstrated that, although YOLOv8 consistently outperforms other YOLO models in low-light scenarios, its detection performance still requires further improvement when compared to leading low-light object detection models.

To effectively enhance the detection accuracy of object detection models in low-light environments while simultaneously reducing computational cost, achieving rapid inference and edge deployment, and mitigating the impact of uncertainty in low-quality images, it is essential to minimize information loss and reduce interference from complex backgrounds and scale variations. Realizing both model lightweighting and efficient edge deployment under such challenging illumination conditions represents a critical research direction for subsequent advancements in low-light object detection.

In response to these challenges, a lightweight network—SCL-YOLOv11—is proposed, which takes YOLOv11 [9] as the baseline and integrates the StarNet [10], C3k2-Star, and LSDECD modules [11]. Furthermore, to address the significant decline in detection accuracy often observed when employing lightweight modules, knowledge distillation is adopted to ensure that the designed model not only achieves low cost and rapid inference but also maintains excellent detection performance. The main contributions are as follows:

1) **StarNet Architecture in the YOLOv11 Backbone:** The StarNet network architecture is introduced into the YOLOv11 backbone to achieve efficient feature representation. By mapping low-quality images from low-light scenarios into a high-dimensional nonlinear feature space, the proposed model significantly enhances shallow feature extraction without increasing computational complexity.

2) **Neck Refinement with Star Blocks:** Star Blocks from the StarNet framework are employed to improve the C3k2 module in the neck, thereby further reducing the computational cost while enhancing the localization accuracy of deep features.

3) **MPDIoU for Bounding Box Regression:** The MPDIoU loss function is used to calculate localization loss, mitigating the risk of gradient explosion while improving model performance in complex scenes, particularly under varying illumination and noise conditions. This approach also accelerates model convergence [12].

4) **LSDECD detection Head:** A lightweight detail-enhanced convolution and a shared-convolution detection head are designed to reduce computational costs while improving the model's ability to capture fine-grained details in challenging scenes [13].

5) **Global Distillation for Performance Preservation:** A global distillation strategy is applied to SCL-YOLOv11 to further improve detection accuracy [14], ensuring that performance remains robust even after the lightweight design [15].

II. SCL-YOLOV11 NETWORK STRUCTURE DESIGN

A. STARNET STRUCTURE DESIGN

Because low-quality targets have fewer relevant feature cues than regular targets, the model tends to lose critical low-quality target features during early convolution and pooling stages, which can lead to the absence of the targets to be detected. This problem is especially prevalent in low-light environments, where low-quality targets are common. Hence, a key challenge in designing target recognition models is how to mitigate feature loss of low-quality targets during early feature extraction [16].

StarNet, proposed by Microsoft in 2024, is a novel lightweight network whose architectural diagram is shown in Figure 1. It adopts a traditional layered network

structure, divided into four stages, each doubling the number of channels. Convolutional layers are used for downsampling, whereas Star Blocks—incorporating depthwise convolutions—perform feature extraction. After the depthwise convolution, batch normalization replaces layer normalization to improve computational efficiency. The feature processing workflow of the Star Blocks is as follows. First, a depthwise separable convolution operation is applied to the input features, and the resulting output is denoted as x. Second, x is processed through two parallel branches: one branch undergoes convolution followed by normalization, while the other undergoes convolution with an activation function followed by batch normalization. The outputs of these two branches are then multiplied element-wise. Finally, the result from the second step is subjected to another depthwise separable convolution and added element-wise to the original input features F, yielding the final output of the Star Blocks module. This computational process is expressed in Equations (1) and (2) [17].

$$x = f_{DWC}(F) \tag{1}$$

$$y = F + f_{DWC}\left(f_C\left(f_R(x) * f_C(x)\right)\right) \tag{2}$$

where **F** is the input feature, is the output of the initial feature processing, $f_{DWC}(x)$ denotes the depthwise separable convolution operation. Meanwhile, y is the output of the Star Blocks, $f_C(x)$ represents the convolution and batch normalization operations, and $f_R(x)$ denotes the operation comprising a convolution with a ReLU activation function followed by batch normalization.

The core of StarNet lies in its "star operation," which can acquire high-dimensional and nonlinear features in a low-dimensional space, perform the necessary computations, and subsequently generate high-dimensional features. Inspired by StarNet's exceptional lightweight design and information extraction capability, this research proposes the SCL-YOLOv11 model, which leverages the StarNet structure to replace the Conv layers and C3k2 blocks in the backbone of YOLOv11. In addition, Star modules from StarNet are fused with the C3k2 blocks in the YOLOv11 head to form the C3k2Star module. This approach further enhances the model's feature extraction and multi-scale information fusion while reducing parameters and computational cost. The architecture of the StarNet model is illustrated in the Backbone of Figure 1.

B. LSDECD LIGHTWEIGHT DETECTION HEAD DESIGN

In the original YOLOv11 detection head, data normalization is required to accelerate model convergence, improve generalization, and prevent gradient vanishing or explosion. The most commonly used normalization technique is batch normalization (BatchNorm), which offers faster convergence, reduced dependence on initial weights, and lower risk of overfitting. However, BatchNorm is highly sensitive to batch size. If the batch size is too small, the computed mean and variance may not adequately represent the entire data distribution, potentially leading to degraded model performance.

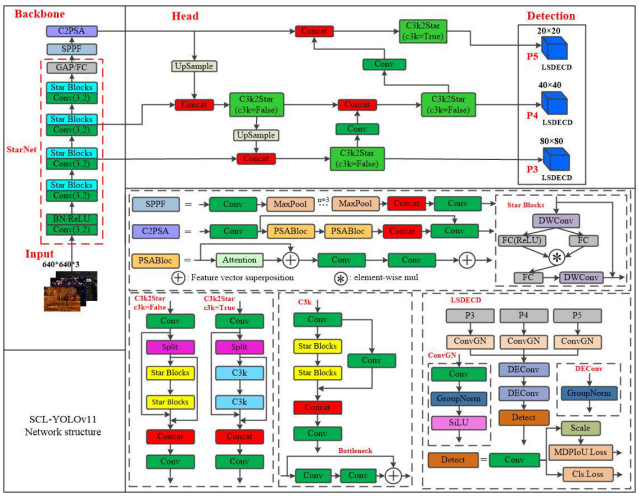


FIGURE 1. SCL-YOLOv11 Network structure.

To effectively address these issues, a lightweight detection head based on detail-enhanced convolution and shared convolution is designed, as depicted in Figure 1. The LSDECD architecture includes the following three components [13]:

1) **Replacing BatchNorm with GroupNorm:** GroupNorm exhibits superior adaptability, preserves the relative relationships among channels, improves model robustness, delivers better performance in certain scenarios, and reduces dependence on batch size.

2) **Employing a Convolutional Weight-Sharing Mechanism:** This mechanism significantly reduces the number of parameters that need to be trained, thereby improving computational efficiency and lowering the risk of overfitting.

3) **Incorporating Detail-Enhanced Convolution (DEConv):** This element amplifies subtle image details, enabling the model to better retain critical features. Reparameterization technology is introduced to achieve an equivalent transformation between DEConv and standard convolution without additional parameters or computational overhead, further enhancing the model's representational and generalization capabilities.

C. LOSS FUNCTION SELECTION IN SCL-YOLOV11

In YOLOv11, the complete intersection over union (CIoU) loss function is employed. Let the anchor box be $B = [x, y, w, h]$ and the target box be $B = [x, y, w, h]$, and let $B_{gt} = [x_{gt}, y_{gt}, w_{gt}, h_{gt}]$. The calculation of CIoU is given by Equations (3)-(7) [12].

$$\text{IoU} = \frac{B_{gt} \cap B}{B_{gt} \cup B} \tag{3}$$

$$L_{\text{CIoU}} = L_{\text{IoU}} + \frac{(x - x_{gt})^2 + (y - y_{gt})^2}{W_g^2 + H_g^2} + \alpha\, v \tag{4}$$

$$L_{\text{IoU}} = 1 - \text{IoU} = 1 - \frac{W_i H_i}{wh + w_{gt} h_{gt} - W_i H_i} \tag{5}$$

$$\alpha = \frac{v}{L_{\text{IoU}} + v} \tag{6}$$

$$v = \frac{4}{\pi^2}\left(\arctan\left(\frac{w}{h}\right) - \arctan\left(\frac{w_{gt}}{h_{gt}}\right)\right)^2 \tag{7}$$

where, W_i, H_i are the overlapping area dimensions of the two boxes, and W_g, H_g represent the dimensions of the smallest enclosing rectangle. The parameter α denotes the weight coefficient, and v is the similarity measure of aspect ratios.

From these definitions, it can be observed that if the bounding boxes share the same width-height ratio but differ in orientation, the aspect-ratio term v in CIoU becomes zero, thereby preventing the model from capturing directional discrepancies in the bounding boxes.

To address this limitation, the proposed approach adopts a new bounding-box similarity measure-Minimum Point Distance-based IoU (MPDIoU)-as the loss function. By calculating the minimum point distances between the upper-left and lower-right corners of the predicted and ground-truth boxes, MPDIoU preserves differences in bounding-box orientation. It not only incorporates factors such as the center distance of overlapping or non-overlapping regions and width-height deviations but also simplifies the computation process. The corresponding formulas are shown in Equations (8)-(10) [18].

$$d_1^2 = (x_1^{prd} - x_1^{gt})^2 + (y_1^{prd} - y_1^{gt})^2 \tag{8}$$

$$d_2^2 = (x_2^{prd} - x_2^{gt})^2 + (y_2^{prd} - y_2^{gt})^2 \tag{9}$$

$$MPDIoU = IoU - \frac{d_1^2}{w^2 + h^2} - \frac{d_2^2}{w^2 + h^2} \tag{10}$$

In Equations (8) and (9), d_1^2 and d_2^2 respectively represent the squared Euclidean distances between the upper-left and lower-right corners of the ground-truth and predicted boxes, reflecting the positional offset between prediction and annotation. The coordinates (x_1^{prd}, y_1^{prd}) and (x_2^{prd}, y_2^{prd}) denote the upper-left and lower-right corners of the predicted bounding box, while (x_1^{gt}, y_1^{gt}) and (x_2^{gt}, y_2^{gt}) denote the corresponding coordinates of the ground-truth bounding box. In Equation (10), IoU represents the intersection over union between the predicted and ground-truth boxes, which can be computed by Equation (3). The terms w and h respectively denote the width and height of the ground-truth bounding box [19].

Summarized from the above equations, MPDIoU takes into account the positional offsets between bounding boxes and simplifies the similarity comparison between predicted and ground-truth boxes. It also remedies the mismatch in object predictions under CIoU, thereby enhancing detection accuracy.

To more intuitively and effectively demonstrate detection performance, the designed network is evaluated via four metrics: Precision (also known as positive predictive value, PPV), Recall (also known as true positive rate, TPR), Average Precision (AP), and mean Average Precision (mAP) [20]. Specifically, mAP@0.5 is the mean average precision at an IoU threshold of 0.5, while mAP@.5:.95 is the average of mAP values computed at IoU thresholds of 0.5, 0.55, 0.6, 0.65, 0.7, 0.75, 0.8, 0.85, 0.9, and 0.95. The calculation formulas for Precision and Recall are shown in Equation (11) and Equation (12).

$$PPV = \frac{TP}{TP + FP}, TPR = \frac{TP}{TP + FN} \tag{11}$$

$$AP = \int_0^1 p(r)dr, mAP = \frac{1}{n}\sum_{k=1}^{n} AP_k \tag{12}$$

where true positive(TP) is the number of instances correctly predicted as positive, false positive(FP) is the number of instances incorrectly predicted as positive, false negative(FN) is the number of instances incorrectly predicted as negative, n is the total number of detection categories, $p(r)$ is the curve plotted with Recall on the x-axis and Precision on the y-axis, which is also considered as the P-R curve, and the enclosed area corresponds to the average precision (AP), hence, K represents the total number of defect categories being considered [21].

III. SIMULATION AND EXPERIMENT

A. EXPERIMENTAL ENVIRONMENT SETUP AND COMPOSITION OF THE EXPERIMENTAL DATASET

The server configuration used in this experiment includes 128 GB of RAM, an NVIDIA GeForce RTX 3090 GPU (24 GB), and an AMD CPU. The deep learning framework employed is PyTorch. The training was conducted for 300 epochs, with 3 warm-up epochs. The initial learning rate was set to 0.01, the weight decay coefficient was 0.0005, the batch size was 16, the momentum in the optimizer was 0.937, and the input image size was 640 × 640.

The SCL-YOLOv11 model uses the ExDark dataset for training, validation, and testing [22]. ExDark is a publicly available dataset specifically designed for object detection and image enhancement under low-light conditions. It contains 7363 low-light images, which were split into training, validation, and testing sets in a 6:2:2 ratio. The dataset covers 12 object categories: Bicycle, Boat, Bottle, Bus, Car, Cat, Chair, Cup, Dog, Motorbike, People, and Table. Sample images from the dataset are displayed in Figure 2.

B. ABLATION EXPERIMENTS

To validate the effectiveness of the modules used in the proposed SCL-YOLOv11 model, eight sets of experiments were designed, each under different conditions. The experimental setup is as follows:

1) Use the original YOLOv11 model as the baseline for the ablation study.

2) Replace the YOLOv11 Backbone with the StarNet structure to enhance extraction of shallow image features without increasing computational complexity.

3) Replace the C3k2 module in the Neck with the C3k2Star module to improve the localization accuracy of deep features.



FIGURE 2. Illustration of ExDark dataset samples.

4) Add the P2 layer as an output for detection to improve performance on small targets [23].

5) Add the P6 layer as an output for detection to improve performance on large targets [24].

6) Use the LSDECD detection head for the output layer to enhance the ability to capture detailed features in complex scenes.

The ablation study demonstrates the influence of each improved module on model performance. Table 1 summarizes the experimental results. Group 1 uses the baseline model (YOLOv11). From Group 2, compared to the baseline model, introducing the StarNet structure in place of the YOLOv11 Backbone increases mAP@0.5 and mAP@.5:.95 by only 0.9% and 0.4%, respectively. However, the parameter count and FLOPs decrease by 26.9% and 20.6%, indicating that the StarNet structure reduces model inference time while retaining adequate detection performance.

Group 4 further shows that employing the C3k2Star module instead of the original C3k2 module in the Neck not only improves detection accuracy but also accelerates inference. Results from Groups 4 and 5 indicate that adding detection layers for small targets (P2) and large targets (P6) does not lead to satisfactory detection performance and, in fact, increases the overall parameter count and FLOPs. Therefore, the final SCL-YOLOv11 design does not incorporate additional detection layers.

Group 6 evaluates the lightweight LSDECD detection head, showing that it significantly enhances detection performance. For instance, mAP@0.5 increases by 0.5%, while mAP@.5:.95 increases by 1.0%. Moreover, the parameter count and FLOPs decrease by 15.4% and 4.8% compared to the original detection head. As a result, the finalized SCL-YOLOv11 model design adopts the StarNet structure, the C3k2Star module, and the LSDECD lightweight detection head as its main components.

To further verify the superiority of the chosen localization loss function, multiple loss functions were compared under the established model framework. The results are shown in Table 2; CIoU was the original loss function in SCL-YOLOv11. Compared to other alternatives, such as DIoU, the proposed MDPIoU function offers obvious improvements in evaluation metrics. Compared to the original CIoU from YOLOv11, MDPIoU increases mAP@0.5 and mAP@.5:.95 by 0.6% and 0.8%, respectively. This improvement occurs because MDPIoU accounts for positional offsets between bounding boxes, simplifies the similarity comparison between prediction and annotation boxes, and addresses the mismatch issue found in the original loss function. Consequently, SCL-YOLOv11 achieves better object detection performance under low-light conditions.

C. ALGORITHM COMPARISON EXPERIMENTS

To further validate the detection performance of the proposed SCL-YOLOv11 model, the original YOLOv11 algorithm was used as a baseline, and experimental comparisons were conducted with three classical object detection algorithms—CenterNet, Faster R-CNN, and SSD—as well as six mainstream algorithms—YOLOv5, YOLOv7, YOLOv8, YOLOv9, YOLOv10, and RT-DETR. In addition, to demonstrate that SCL-YOLOv11 not only achieves excellent detection performance but also offers extremely low computational cost and superior deployment advantages, the GOI-YOLO model (as reported in Reference 6) and the 3L-YOLO model (as reported in Reference 7) were included in the comparative analysis. The experimental results are presented in Table 3.

The comparative experiments indicate that, relative to the other 12 object detection algorithms, the proposed SCL-YOLOv11 model requires only 1.6 million parameters and 4.7 billion FLOPs—the lowest among all compared models. Moreover, the reduction in parameter count does not lead to a dramatic deterioration in detection performance; the model still maintains satisfactory detection accuracy. Although SCL-YOLOv11 exhibits significant advantages in terms of parameter count and FLOPs compared to the YOLO series and RT-DETR models, its detection performance metrics show a certain degree of decline. In particular, when compared with the latest GOI-YOLO model, the mAP@0.5 and mAP@.5:.95 values of SCL-YOLOv11 decrease by 6.7% and 3.9%, respectively. Therefore, further enhancement of the detection accuracy of SCL-YOLOv11 remains an urgent issue.

D. KNOWLEDGE DISTILLATION FOR SCL-YOLOV11

Although the proposed SCL-YOLOv11 model demonstrates satisfactory detection performance on the ExDark dataset, a notable issue remains: as the number of parameters and network layers decreases, detection accuracy tends to decline. To address this limitation and further enhance detection performance, three types of knowledge distillation experiments were conducted—Feature Distillation, Logical Distillation, and Feature and Logical Distillation.

In the knowledge distillation experiments, the SCL-YOLOv11 model was designated as the student model, while the YOLOv8 model trained in this study served as the

TABLE 1. SCL-YOLOv11 network structure of ablation experiment contrast.

Group No.	StarNet	C3k2Star	P2 layer	P6 layer	LSDECD	Parameter (M)	FLOPs (640@B)	mAP@0.5 (%)	mAP@.5:.95 (%)
1	-	-	-	-	-	2.6	6.3	63.9	39.4
2	✓	-	-	-	-	1.9	5.0	64.8	39.8
3	-	✓	-	-	-	2.5	6.3	64.9	39.8
4	-	-	✓	-	-	2.7	10.4	62.1	37.1
5	-	-	-	✓	-	4.0	6.4	64.3	39.2
6	-	-	-	-	✓	2.2	6.0	64.4	40.4

TABLE 2. SCL-YOLOv11 network uses different loss functions for comparison.

Evaluation Index	CIoU Loss	DIoU Loss	EIoU Loss	GIoU Loss	WIoU Loss	PIoU Loss	SIoU Loss	MDPIoU Loss
mAP@0.5(%)	63.9	64.1	64.2	64.4	64.3	64.5	64.1	64.5
mAP@.5:.95(%)	39.4	39.4	39.5	39.1	39.6	39.3	39.5	40.2

TABLE 3. Comparison of detection results of different algorithms on test dataset.

Model Types	Pretrained weights	Parameter (M)	FLOPs (640@B)	mAP@0.5 (%)	mAP@.5:.95 (%)
YOLOv11 [9]	YOLOv11n	2.6	6.3	63.9	39.4
YOLOv10 [25]	YOLOv10n	8.1	24.8	64.4	40.1
YOLOv9 [26]	YOLOv9n	9.9	40.6	65.3	40.8
YOLOv8 [27]	YOLOv8n	3.0	8.1	68.4	42.6
YOLOv7 [28]	YOLOv7n	37.2	105.2	66.3	41.3
YOLOv5 [29]	YOLOv5n	7.3	17.0	64.9	40.2
RT-DETR [30]	rtdetr_r18vd (backbone=resnet18)	20.0	57.0	67.6	41.4
GOI-YOLO [6]	YOLOv8n	2.7	5.9	**71.2**	**44.1**
3L-YOLO [7]	YOLOv8n	5.9	16.6	68.8	42.0
SSD [31]	SSD300	26.3	62.7	45.8	29.5
Faster R-CNN [32]	FasterRCNN (backbone=resnet50)	137.1	370.2	37.1	18.7
CenterNet [33]	CenterNet (backbone=resnet50)	32.7	70.2	39.4	20.1
SCL-YOLOv11	**YOLOv11n**	**1.6**	**4.7**	64.5	40.2

teacher model. The distillation temperature parameter was set within the range T = 20, 25, 30, 35, 40, and the effects of varying temperatures on model performance were compared to determine the optimal distillation settings. As indicated in Table 4, detection accuracy increased gradually with rising distillation temperatures; however, beyond a certain threshold, accuracy began to decline, exhibiting an overall trend of initial improvement followed by deterioration.

The SCL-YOLOv11 model achieved optimal performance under the Feature and Logical Distillation approach with a distillation temperature of T = 30. Under these conditions, the distilled model attained mAP@0.5 and mAP@.5:.95 values of 67.6% and 42.4%, respectively, representing improvements of 3.1% and 2.2% compared to the pre-distillation performance. These results confirm that the application of knowledge distillation significantly enhances the detection performance of the SCL-YOLOv11 model.

To further verify that the final SCL-YOLOv11 model surpasses YOLOv8 and YOLOv11 in inference speed, frames per second (FPS) were measured (see Equation (13)), FPS represents the number of images processed by the model per second. During the test of inference speed, 200 images were designated for model warming, and the inference time

TABLE 4. Experimental results of global pruning and knowledge distillation of the model.

Distillation Types	Distillation Temperature	mAP@0.5(%)	mAP@.5:.95(%)
Feature Distillation	20	65.2	38.1
	25	**66.7**	**41.2**
	30	66.2	39.3
	35	65.5	38.6
	40	65.0	38.1
Logical Distillation	20	65.5	38.6
	25	**66.3**	**40.7**
	30	65.8	39.5
	35	65.1	38.7
	40	64.8	38.2
Feature And Logical Distillation	20	66.2	39.8
	25	67.1	41.5
	30	**67.6**	**42.4**
	35	67.3	41.7
	40	65.4	39.1

for one image was measured under consistent experimental conditions. Table 5 shows that the FPS of SCL-YOLOv11 increases by 18.8% and 17.5% compared to YOLOv8 and YOLOv11, respectively. This outcome indicates that SCL-YOLOv11 offers both robust detection performance and high inference speed.

$$FPS = \frac{1000}{T_{pre} + T_i + T_{post}} \tag{13}$$

where, FPS is in frames per second (f/s), T_{pre} is the preprocessing time, T_i is the inference time, and T_{post} is the postprocessing time per image.

TABLE 5. Comparison of FPS of three models.

Model Types	Device	Warmup Figures	Model Size (MByte)	FPS
YOLOv8	NVIDIA GeForce		6.0	93.4
YOLOv11	RTX 3090	200	5.4	94.5
SCL-YOLOv11	(24575MiB)		3.6	111.0

E. MODEL PERFORMANCE TESTING

To effectively validate the detection performance of the final SCL-YOLOv11 model on edge devices, a new ExDark-Test-Datasets collection was formed from 128 low-light images not used in training. Figure 3 displays some sample images from this test dataset. The resulting data reveal that YOLOv11 missed seven targets under low-light conditions and complex backgrounds (as shown by the red boxes in Figure 4(b)), whereas SCL-YOLOv11 missed only two targets. In addition, YOLOv11 produced one false detection (yellow box in Figure 4(b)), whereas SCL-YOLOv11 did not exhibit any false detections. Lastly, Figures 4(a) and 4(b) show that the heatmap for SCL-YOLOv11 is more focused on the actual target, indicating less attention to the background. This behavior demonstrates that SCL-YOLOv11 achieves stronger robustness and higher detection accuracy in complex low-light environments.



FIGURE 3. ExDark-Test-Datasets image dataset.

The final test results on the testing dataset are presented in Table 6. The experimental outcomes indicate that the SCL-YOLOv11 model achieves the best recognition performance



FIGURE 4. Detection results of YOLOv11 and SCL-YOLOv11 on the ExDark-Test-Datasets dataset.

for large-scale objects—such as the "Bus" category—in the ExDark-Test-Datasets, with AP@0.5 and AP@.5:.95 values reaching 84.1% and 65.2%, respectively. In contrast, detection performance for small-scale objects, such as those in the "Cup" and "Cat" categories, is only moderate. Furthermore, the model exhibits the poorest performance when detecting objects like "Chair" and "Table" within complex low-light backgrounds. Consequently, enhancing detection accuracy for targets similar to "Chair" and "Table" in complex environments, as well as further optimizing the detection of small-scale objects, represents a key direction for future improvements of the SCL-YOLOv11 model.

TABLE 6. Detection accuracy for each category in the ExDark-Test-Datasets by the SCL-YOLOv11 Model.

Object Types	Precision (%)	Recall (%)	AP@0.5 (%)	AP@.5:.95 (%)
Bicycle	86.8	69.1	79.2	52.4
Boat	73.4	59.5	67.8	36.5
Bottle	82.5	53.4	65.0	40.7
Bus	**83.5**	**77.8**	**84.1**	**65.2**
Car	75.3	62.4	70.0	45.2
Cat	68.0	59.5	65.6	39.6
Chair	63.4	51.5	55.7	33.7
Cup	76.7	51.1	62.5	37.6
Dog	79.6	61.7	72.7	47.5
Motorbike	66.0	59.7	66.4	40.7
People	79.0	58.6	69.6	37.8
Table	65.5	46.5	52.7	31.8

F. VERIFICATION OF THE GENERALIZATION CAPABILITY OF THE SCL-YOLOV11 MODEL

To validate the robustness of the SCL-YOLOv11 model in a broader range of low-light scenarios, images captured under low-light conditions were selected from the PASCAL VOC dataset for generalization testing. A specialized subset, designated as the PASCAL_VOC_Dark dataset, was compiled

and consists of 3,500 images. These images randomly include 20 target categories: "aeroplane," "bicycle," "bird," "boat," "bottle," "bus," "car," "cat," "chair," "cow," "diningtable," "dog," "horse," "motorbike," "person," "pottedplant," "sheep," "sofa," "train," and "tvmonitor." A sample illustration of the PASCAL VOC dataset is shown in Figure 5. The dataset was partitioned into training, validation, and testing sets at a ratio of 6:2:2.



FIGURE 5. Sample illustrations of the PASCAL_VOC_Dark dataset.

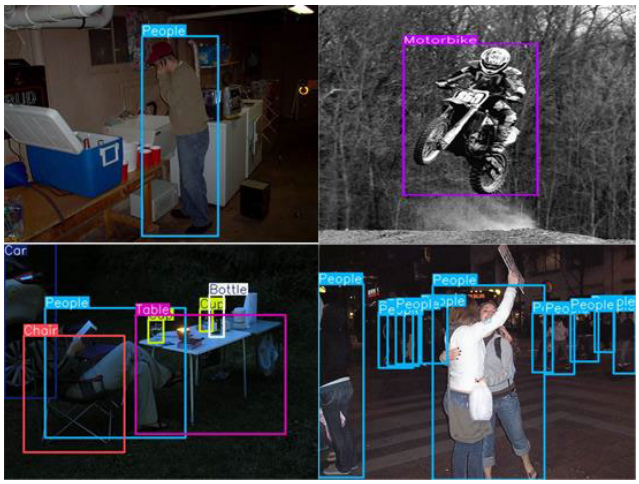


FIGURE 6. Detection results of SCL-YOLOv11 on the PASCAL_VOC_Dark dataset.

TABLE 7. Overall detection performance of two models on the PASCAL_VOC_Dark dataset.

Network Model	Parameter (M)	FLOPs (640@B)	mAP@0.5 (%)	mAP@.5:.95 (%)
YOLOv8	3.0	8.1	54.0	41.2
YOLOv11	2.6	6.3	51.2	37.8
SCL-YOLOv11	1.6	4.7	52.8	38.5

The detection performance of the SCL-YOLOv11 model on the PASCAL_VOC_Dark dataset is presented in Figure 6 and Table 7. According to the experimental findings, compared to YOLOv11, the SCL-YOLOv11 model not only maintains robust inference speed but also achieves improvements of 1.6% and 0.7% in mAP@0.5(%) and mAP@.5:.95(%), respectively. Furthermore, when compared to the YOLOv8 model, although the mAP@0.5(%) and mAP@.5:.95(%) values of SCL-YOLOv11 decrease by 1.2% and 2.7%, respectively, its parameter count and FLOPs are only 53.3% and 58.0% of those of YOLOv8. These results indicate that the SCL-YOLOv11 model can achieve excellent detection performance on the PASCAL_VOC_Dark dataset with minimal computational cost.

IV. CONCLUSION

To address issues such as heavy image noise, blurred edges, and loss of detail in detection tasks under complex low-light conditions, a lightweight target recognition network termed SCL-YOLOv11 is introduced. In response to the limitations of the original YOLOv11 model—namely poor recognition performance and prolonged inference time under low-light conditions—SCL-YOLOv11 incorporates the following key enhancements. First, the StarNet architecture is integrated into the YOLOv11 backbone to improve the extraction of shallow image features without increasing computational complexity. Next, Star Blocks from the StarNet framework are used to optimize the C3k2 module in the Neck, forming the C3k2Star module to enhance the localization accuracy of deep features. Additionally, the MPDIoU loss function replaces the original CIoU, effectively preventing gradient explosion. Subsequently, the LSDECD lightweight detection head is designed to further improve the network's ability to capture fine-grained details in complex scenarios. Finally, global distillation is employed to complete knowledge distillation for the SCL-YOLOv11 model, thereby preserving detection performance after lightweight optimization. The resulting model offers an efficient, accurate, and resource-efficient detection framework, facilitating industrial deployments on edge devices and further advancing object detection technology in low-light environments.

ACKNOWLEDGMENT

The authors would like to thank Hainan Vocational University of Science and Technology.

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